

Riley Beenders

SOLUTIONS ARCHITECTURE | TECHNOLOGY STRATEGY | PLATFORM STANDARDIZATION

Bethlehem, PA | [\(484\) 523-4431](tel:(484)523-4431) | rileybeenders@gmail.com | www.rileybeenders.com | linkedin.com/in/riley-beenders

PROFESSIONAL SUMMARY

Technology and engineering leader with 10+ years of progressive multidisciplinary experience, including CTO and Lead R&D roles. Establishes standardized architectures across hardware, software, and enterprise platforms, leads post-acquisition integration and standardization, and partners with vendors, global teams, and business stakeholders to deliver scalable, secure, and resilient systems. Evaluates emerging technology, including AI-assisted diagnostics and local LLM tooling, and balances business needs, engineering constraints, cost, and risk in architecture decisions.

CORE EXPERTISE

Architecture & Strategy: Solution and system architecture; platform standardization; technology strategy; architecture decisions balancing cost, reliability, and scalability; risk analysis and mitigations; emerging-technology evaluation; proof-of-concept research

Platforms & Integration: Kubernetes; Docker; Proxmox virtualization; Windows/Linux administration; ERP/CRM development and Microsoft Business Central integration; Modbus and CAN-bus integration; remote operations platforms

Software & Emerging Tech: TypeScript/React/Next.js (Node.js runtime); Python; PowerShell; LabVIEW; Git/GitHub and GitHub Actions; AI-assisted diagnostics; local LLM tooling (LM Studio, oMLX)

Leadership & Delivery: Cross-functional and worldwide collaboration; hiring and technical interviews; vendor and third-party partner management; cost negotiation; requirements definition; technical documentation

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Re-engineered POSEIDON and ICARUS with a standardized linear-motor architecture and swappable mounts, replacing servo-driven ballscrews, demonstrating 4-micron precision at 4 m/s with a 30 kg moving mass, and reducing support complexity.
- Managed and expanded the division's IT infrastructure after the Proteor acquisition, deploying cloud gateways and standardizing software and internal processes to align with company-wide systems.
- Managed the Coplay site's Microsoft Business Central ERP rollout, syncing SKUs with Proteor's global locations.
- Implemented ProteorPrint OS with external software partner SimplyPrint, contributing machine requirements and integration for remote operation, live video, AI-assisted diagnostics, and material, waste, and outcome metrics.
- Built an automated LabVIEW QC system that negotiates Modbus communication with three tri-axial laser sensors over a dedicated private VLAN, collecting nine data points (three per laser) every 17.5 ms and charting live tolerances.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.
- Led the design and development of ICARUS-Lite, owning its mechanical, electrical, and software design, including custom AC heaters and an IEEE-style electrical design adaptable to 120/240 VAC.
- Owned ICARUS-Lite through commercialization with plug-and-play setup, contributing to increased revenue.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer

January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM for orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing.

- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy and conducted early 6-axis robotic non-planar printing research.
- Led the design and launch of Gen3 POSEIDON, combining a 1 m³ build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives that prevent uncontrolled motion during power loss, industrial-servo XY motion, and a custom gantry.

Lead Electro-Mechanical Engineer

June 2022 - January 2023

- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging while supporting larger, modular machines.
- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.

Industrial Engineer

April 2021 - June 2022

- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry system for a large-scale statue client, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.

Engineering Technician

May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.

SELECTED PROJECTS

Engineering Operations Platform - Kubernetes ERP/CRM | Filament Innovations | Redundant Kubernetes-hosted system connecting engineering design data to operations: centralized orders, inventory, tasks, projects, and customer interactions, and linked dynamic BOM generation to purchasing.

PROTEOR Print Integration - Remote Operations Software | Proteor | Single interface for remote operation and diagnosis from any networked device, delivered with an external software partner: live video monitoring, machine-vision AI-assisted failure detection and alerts, and tracking of material usage, waste, and print success or failure for analytics-driven decisions.

RileyBeenders.com - Full-Stack Portfolio Platform | 2026 | Built and deployed a TypeScript/React/Next.js site on Vercel with structured JSON content, a Node.js PDF endpoint, Git version control, GitHub Actions automation on every push to main, and web analytics.

Linear Motor Platform Standardization | Proteor | Converted both flagship platforms from servo-driven ballscrews to a shared linear-motor architecture with swappable mounts, and validated 4-micron precision at 4 m/s with a 30 kg moving mass on the machine platforms.

Modular CAN-Bus Controls Architecture | Filament Innovations | Self-contained, independently testable electronics enclosures connected over CAN bus, separating electrical build work from final machine integration and cutting installation of tested electronics to minutes.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed 750+ wireless access points across six school buildings; automated setup-data entry for 2,500+ Chromebooks with an Arduino-based tool; coordinated distribution with school administrators.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online store for printer heat inserts and accessories; produced plastic face shields during the COVID-19 pandemic.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Certifications: MathWorks MATLAB, Simulink, Simscape, and Machine Learning Onramps (2026); Lean Six Sigma Yellow Belt and White Belt, Educate 360 (2025); Penn State Certificate in Engineering Design and Tools (2023); Lafayette College Mechanical Engineering Certificate (2019)